

Oral Exam

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September 2025

1 History

Over my time at Utah so far, the following topics are the most relevant subjects I have studied. In chronological order:

- Macaulay2 and Resolving Singularities on Surfaces (following Karl's Macaulay2 seminar)
- Complex Geometry (following Harold's class, from Huybrechts's *Complex Geometry* Chapters 1-5)
- Syzygies (Ein-Lazarsfeld *Lectures on the Syzygies and Geometry of Algebraic Varieties* 1-6), in particular:
 - Minimal Resolutions
 - Castelnuovo-Mumford Regularity
- Derived/Triangulated Categories (in algebraic geometry or otherwise)
 - Definitions (Gelfand Manin *Methods in Homological Algebra* II-IV)
 - Exceptional Collections/Tilting Bundles (Calabrese's *On a Theorem by Beilinson*, Craw's *Explicit Methods for Derived Categories of Sheaves* Lectures 1-2)
 - Quiver Representations (Assem, Simson, Skowronski's *Elements of the Representation Theory of Associative Algebras* Chapters I-III)
 - In Algebraic Geometry (Huybrechts *FM Transforms* 1-8)
 - t-Structures and Bridgeland Stability Conditions (Bayer's *A tour to stability conditions on derived categories*)
- Schemes (following Harold's class, Hartshorne II.1-II.8, IV.1)
- Representation Theory of Complex Lie Algebras (from Dragan's class, and Fulton and Harris *Representation Theory* Chapters 14, 15, 18)
- Cohomology Base Change (Vakil Chapters 24-25)

- Construction of the Hilbert Scheme (Alper’s *Stacks and Moduli* Chapter 1, Mumford’s *Lectures on Curves and Surfaces* for Flattening Stratification)
- Theorem on Formal Functions (Vakil Chapter 28) and applications to Derived Categories (Bondal’s *Projective Bundles, Monoidal Transformations, and Derived Categories of Coherent Sheaves*)

2 Interests

I have particularly enjoyed understanding the derived categories of Fano varieties through exceptional collections/semiorthogonal decompositions. I spent a lot of time wrestling with the derived equivalence of $\mathrm{Coh}(\mathbb{P}^n)$ and representations of “the” Beilinson quiver (you get different options for the two different Beilinson collections, or any mutations).

I then moved on to study Homological Projective Duality, in particular to understand the relation between pencils of quadrics and hyperelliptic curves from a derived category perspective. To start, I needed to understand the derived category of just one quadric by Kapranov, which took me on a journey through Koszul rings (to generalize Beilinson’s resolution of the diagonal) and representation theory (to understand the cohomology of “spinor bundles” by Borel-Weil-Bott). There are still many gaps in my understanding of this story, but I am thoroughly enjoying it.

I have also started trying to think about moduli. I am starting out with the Hilbert scheme, but I have my sights set also on moduli spaces of semistable complexes (with a fixed Chern character) with respect to a given Bridgeland stability condition, and also moduli spaces appearing in Quantum cohomology.

3 Syllabus

I hope to spend a lot of my oral exam discussing my most recent focus on the construction of the Hilbert scheme and necessary cohomology results (like “Cohomology and Base Change” theorems) to describe the morphism to the appropriate Grassmannian. I am also particularly fond of Beilinson’s resolution of the diagonal.

I am then happy to talk about foundational algebraic geometry, derived categories, or representation theory—and then to continue wherever our discussion goes.